

2021 AIME II Problems/Problem 12

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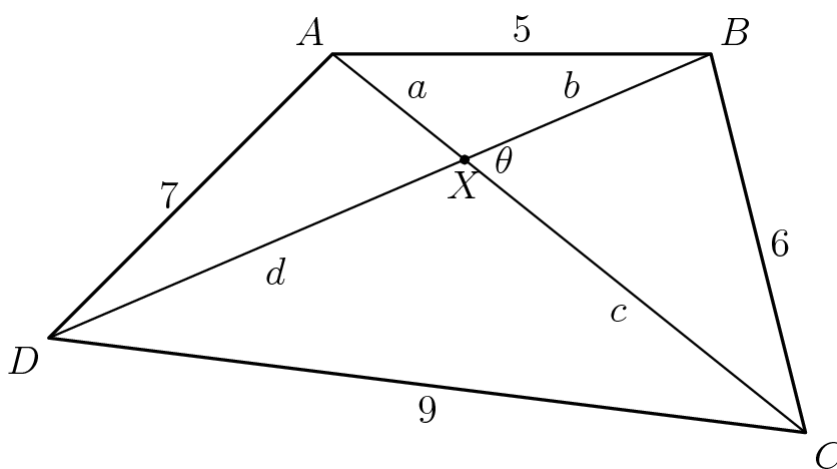
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Problem

A convex quadrilateral has area 30 and side lengths 5, 6, 9, and 7, in that order. Denote by θ the measure of the acute angle formed by the diagonals of the quadrilateral. Then $\tan \theta$ can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Solution 1 (Sines and Cosines)

Since we are asked to find $\tan \theta$, we can find $\sin \theta$ and $\cos \theta$ separately and use their values to get $\tan \theta$. We can start by drawing a diagram. Let the vertices of the quadrilateral be A , B , C , and D . Let $AB = 5$, $BC = 6$, $CD = 9$, and $DA = 7$. Let $AX = a$, $BX = b$, $CX = c$, and $DX = d$. We know that θ is the acute angle formed between the intersection of the diagonals AC and BD .



We are given that the area of quadrilateral $ABCD$ is 30. We can express this area using the areas of triangles AXB , BXC , CXD , and DXA . Since we want to find $\sin \theta$ and $\cos \theta$, we can represent these areas using $\sin \theta$ as follows:

$$30 = [ABCD] \\ = [AXB] + [BXC] + [CXD] + [DXA]$$

$$\begin{aligned}
&= [AAB] + [BAC] + [CAD] + [DAA] \\
&= \frac{1}{2}ab \sin(\angle AXB) + \frac{1}{2}bc \sin(\angle BXC) + \frac{1}{2}cd \sin(\angle CXD) + \frac{1}{2}da \sin(\angle AXD) \\
&= \frac{1}{2}ab \sin(180^\circ - \theta) + \frac{1}{2}bc \sin(\theta) + \frac{1}{2}cd \sin(180^\circ - \theta) + \frac{1}{2}da \sin(\theta).
\end{aligned}$$

We know that $\sin(180^\circ - \theta) = \sin \theta$. Therefore it follows that:

$$\begin{aligned}
30 &= \frac{1}{2}ab \sin(180^\circ - \theta) + \frac{1}{2}bc \sin(\theta) + \frac{1}{2}cd \sin(180^\circ - \theta) + \frac{1}{2}da \sin(\theta) \\
&= \frac{1}{2}ab \sin(\theta) + \frac{1}{2}bc \sin(\theta) + \frac{1}{2}cd \sin(\theta) + \frac{1}{2}da \sin(\theta) \\
&= \frac{1}{2} \sin \theta (ab + bc + cd + da).
\end{aligned}$$

From here we see that $\sin \theta = \frac{60}{ab + bc + cd + da}$. Now we need to find $\cos \theta$. Using the Law of Cosines on each of the four smaller triangles, we get following equations:

$$\begin{aligned}
5^2 &= a^2 + b^2 - 2ab \cos(180^\circ - \theta), \\
6^2 &= b^2 + c^2 - 2bc \cos \theta, \\
9^2 &= c^2 + d^2 - 2cd \cos(180^\circ - \theta), \\
7^2 &= d^2 + a^2 - 2da \cos \theta.
\end{aligned}$$

We know that $\cos(180^\circ - \theta) = -\cos \theta$ for all θ . We can substitute this value into our equations to get:

$$5^2 = a^2 + b^2 + 2ab \cos \theta, \quad (1)$$

$$6^2 = b^2 + c^2 - 2bc \cos \theta, \quad (2)$$

$$9^2 = c^2 + d^2 + 2cd \cos \theta, \quad (3)$$

$$7^2 = d^2 + a^2 - 2da \cos \theta. \quad (4)$$

If we subtract $(2) + (4)$ from $(1) + (3)$, the squared terms cancel, leaving us with:

$$\begin{aligned}
5^2 + 9^2 - 6^2 - 7^2 &= 2ab \cos \theta + 2bc \cos \theta + 2cd \cos \theta + 2da \cos \theta \\
21 &= 2 \cos \theta (ab + bc + cd + da).
\end{aligned}$$

From here we see that $\cos \theta = \frac{21/2}{ab + bc + cd + da}$.

Since we have figured out $\sin \theta$ and $\cos \theta$, we can calculate $\tan \theta$.

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\frac{60}{ab+bc+cd+da}}{\frac{21/2}{ab+bc+cd+da}} = \frac{60}{21/2} = \frac{120}{21} = \frac{40}{7}.$$

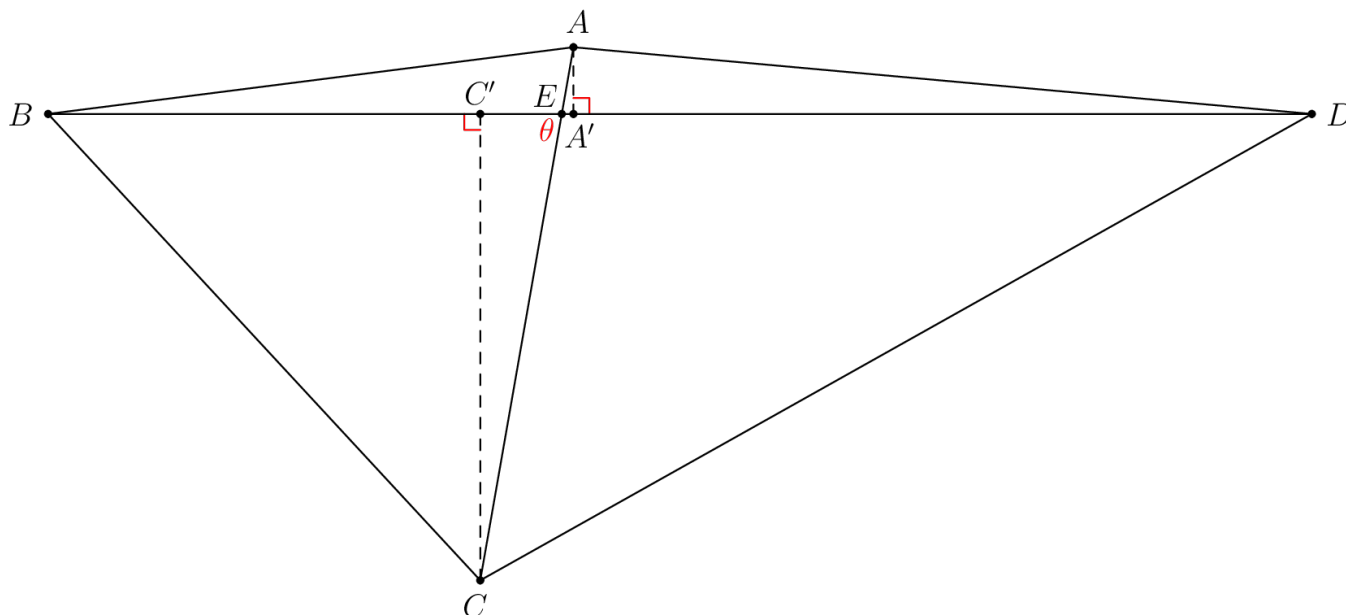
Therefore our answer is $40 + 7 = \boxed{047}$.

~ Steven Chen (www.professorchen.edu.com)

~ my_aops_lessons

Solution 2 (Right Triangles)

In convex quadrilateral $ABCD$, let $AB = 5$, $BC = 6$, $CD = 9$, and $DA = 7$. Let A' and C' be the feet of the perpendiculars from A and C , respectively, to $\overline{BD}$. We obtain the following diagram:



Let $BC' = p$, $C'E = q$, $EA' = r$, $A'D = s$, $AA' = h_1$, and $CC' = h_2$. We apply the Pythagorean Theorem to right triangles $\triangle ABA'$, $\triangle BCC'$, $\triangle CDC'$, and $\triangle DAA'$, respectively:

$$(p + q + r)^2 + h_1^2 = 5^2, \quad (1)$$

$$p^2 + h_2^2 = 6^2, \quad (2)$$

$$(q + r + s)^2 + h_2^2 = 9^2, \quad (3)$$

$$s^2 + h_1^2 = 7^2. \quad (4)$$

Let the brackets denote areas. We get

$$[ABD] + [CBD] = [ABCD]$$

$$\frac{1}{2}(p + q + r + s)h_1 + \frac{1}{2}(p + q + r + s)h_2 = 30$$

$$\frac{1}{2}(p + q + r + s)(h_1 + h_2) = 30$$

$$(p + q + r + s)(h_1 + h_2) = 60. \quad (5)$$

We subtract (2) + (4) from (1) + (3) :

$$(p + q + r)^2 + (q + r + s)^2 - p^2 - s^2 = 21$$

$$[(p + q + r)^2 - s^2] + [(q + r + s)^2 - p^2] = 21$$

$$(p + q + r + s)(p + q + r - s) + (p + q + r + s)(-p + q + r + s) = 21$$

$$(p + q + r + s)(2q + 2r) = 21$$

$$2(p + q + r + s)(q + r) = 21$$

$$(p + q + r + s)(q + r) = \frac{21}{2}. \quad (6)$$

From right triangles $\triangle AEA'$ and $\triangle CEC'$, we have $\tan \theta = \frac{h_1}{r} = \frac{h_2}{q}$. It follows that

$$\tan \theta = \frac{h_1}{r} \implies h_1 = r \tan \theta, \quad (1\star)$$

$$\tan \theta = \frac{h_2}{q} \implies h_2 = q \tan \theta. \quad (2\star)$$

Finally, we divide (5) by (6) :

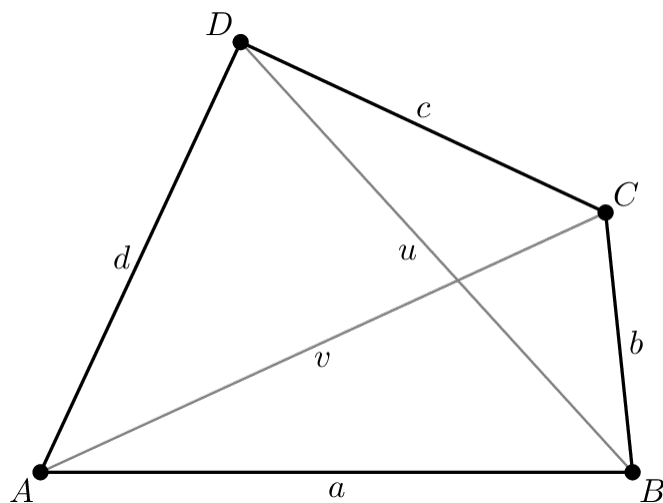
$$\begin{aligned} \frac{h_1 + h_2}{q + r} &= \frac{40}{7} \\ \frac{r \tan \theta + q \tan \theta}{q + r} &= \frac{40}{7} && \text{by (1}\star) \text{ and (2}\star) \\ \frac{(r + q) \tan \theta}{q + r} &= \frac{40}{7} \\ \tan \theta &= \frac{40}{7}, \end{aligned}$$

from which the answer is $40 + 7 = \boxed{047}$.

~MRENTHUSIASM

Solution 3 (Bretschneider's Formula)

Bretschneider's Formula



Given quadrilateral $ABCD$, let, a, b, c, d , be the sides, s the semiperimeter, and u, v , the diagonals. Then the area, K , is given by

$$K = \frac{1}{4} \sqrt{4u^2v^2 - (b^2 + d^2 - a^2 - c^2)^2}$$

Solution

By Bretschneider's Formula,

$$30 = \frac{1}{4} \sqrt{4u^2v^2 - (b^2 + d^2 - a^2 - c^2)^2} = \frac{1}{4} \sqrt{4u^2v^2 - 441}.$$

Thus, $uv = \frac{3\sqrt{1649}}{2}$. Also,

$$[ABCD] = \frac{1}{2} \cdot uv \sin \theta;$$

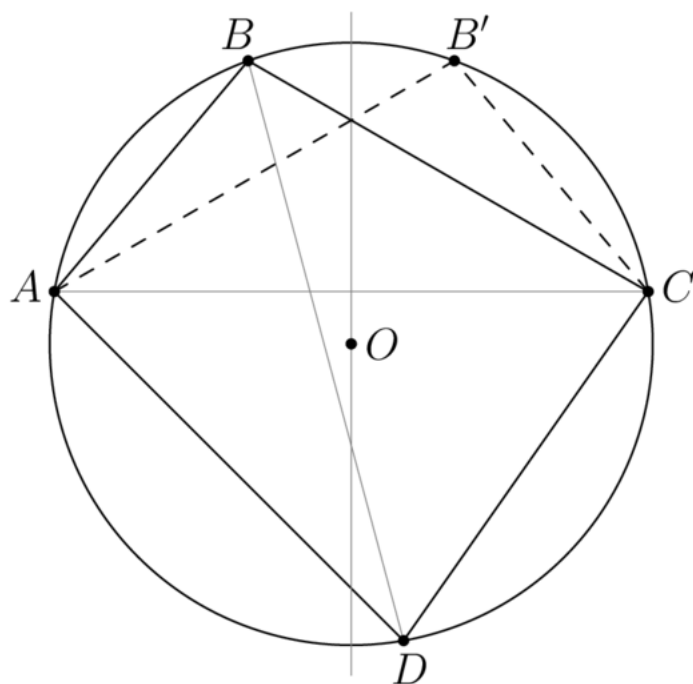
solving for $\sin \theta$ yields $\sin \theta = \frac{40}{\sqrt{1649}}$. Since θ is acute, $\cos \theta$ is positive, from which $\cos \theta = \frac{7}{\sqrt{1649}}$. Solving for $\tan \theta$ yields

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{40}{7},$$

for a final answer of 047.

~ Leo.Euler

Solution 4 (Symmetry)



Claim

Given an inscribed quadrilateral $ABCD$ with sides $AB = a$, $BC = b$, $CD = c$, and $DA = d$. Prove that the $\angle \theta < 90^\circ$ between the diagonals is given by

$$2(ac + bd) \cos \theta = |d^2 - c^2 + b^2 - a^2|.$$

Proof

Let the point B' be symmetric to B with respect to the perpendicular bisector AC . Then the quadrilateral $AB'CD$ is an inscribed one, $AB' = b$, $B'C = a$.

$$2\angle AEB = \widehat{AB} + \widehat{CD}.$$

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$$2\angle B'AD = B'C + CD = AB + CD \implies \angle AEB = \angle B'AD.$$

We apply the Law of Cosines to $\triangle AB'D$ and $\triangle CB'D$:

$$B'D^2 = AD^2 + AB'^2 - 2AD \cdot AB' \cos \theta,$$

$$B'D^2 = CD^2 + CB'^2 + 2CD \cdot CB' \cos \theta,$$

$$d^2 + b^2 - 2bd \cos \theta = c^2 + a^2 + 2ac \cos \theta,$$

$$2(ac + bd) \cos \theta = |d^2 - c^2 + b^2 - a^2|.$$

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Note 1

By generalization, the tangent of the acute angle formed by the diagonals is

$$\left| \frac{4A}{a^2 - b^2 + c^2 - d^2} \right|.$$

Video Solution by MOP 2024

<https://youtu.be/K0u0ACMTSw8>

~r00tsOfUnity

Video Solution

<https://www.youtube.com/watch?v=7DxldTLNbo0>

Video Solution by Interstigation

<https://youtu.be/8GR04za5rPI>

~Interstigation

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